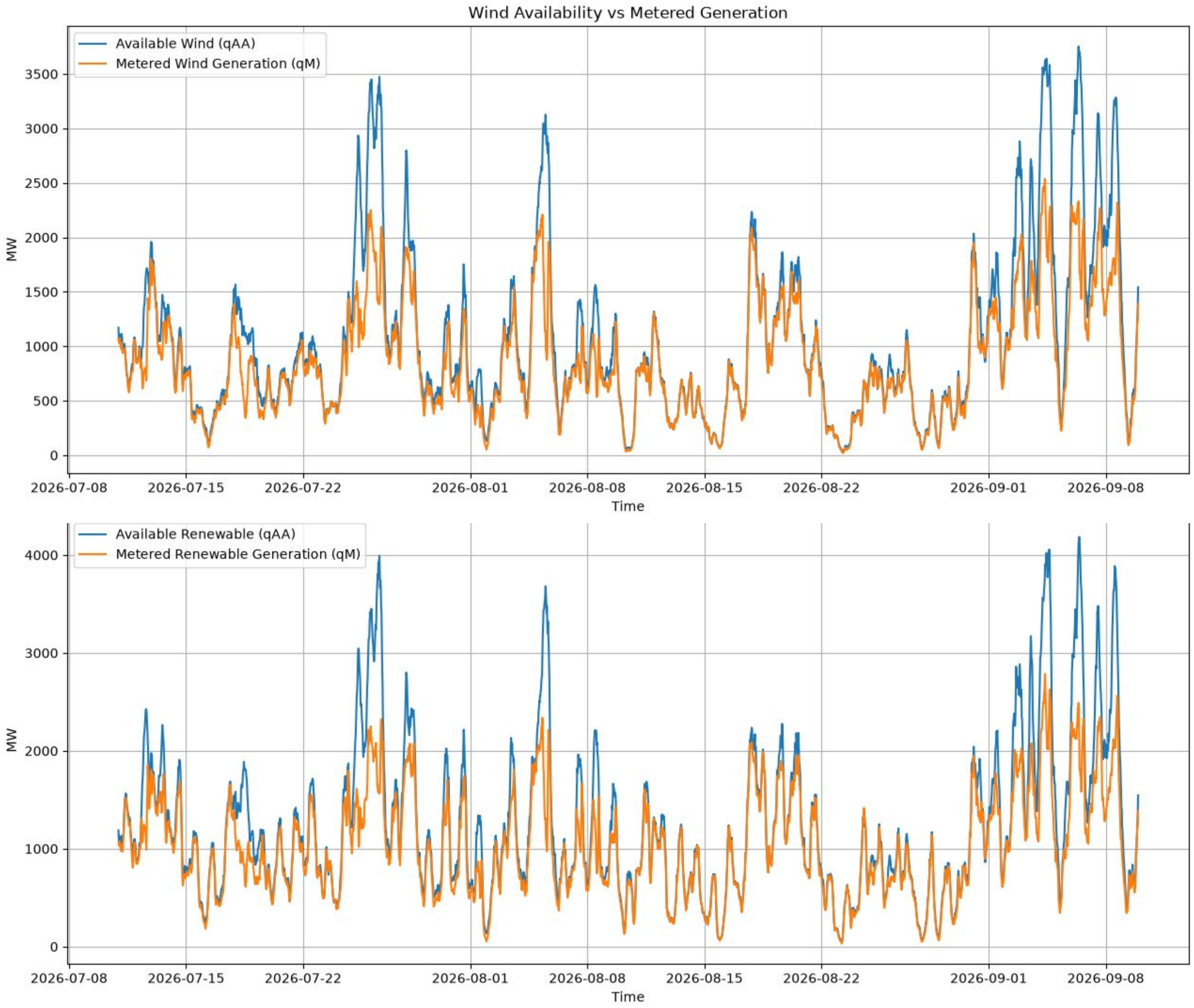
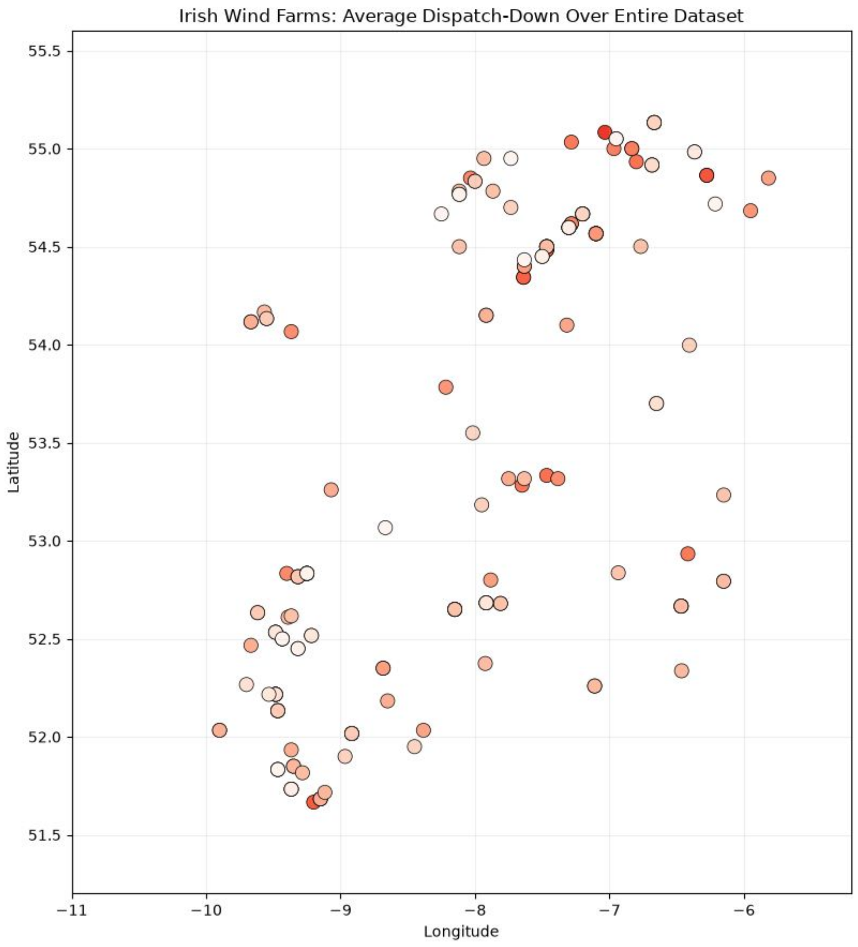
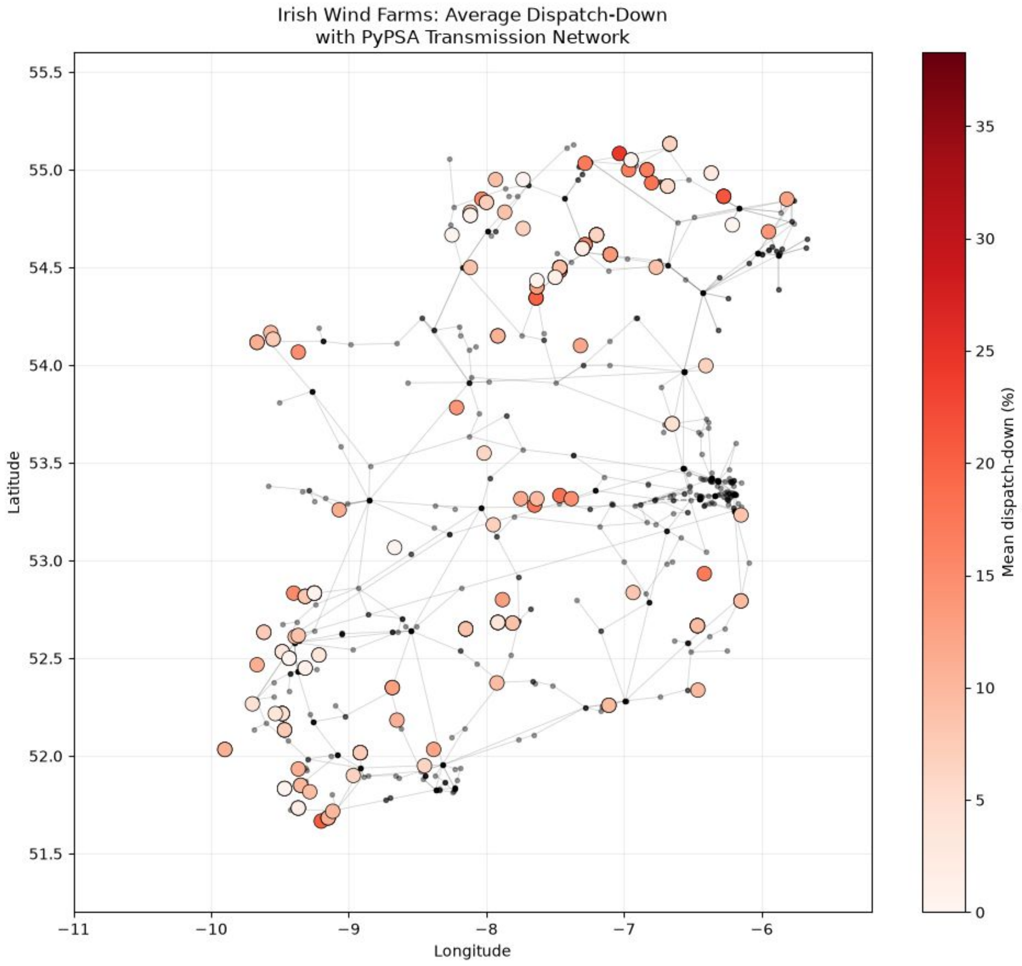
Tackling the problem of constraint group construction to reduce wind energy waste and defend against cascading outages

Team 6


Wind Availability vs Metered Generation
Available Wind (qAA)
Metered Wind Generation (qM)
MW
0
500
1000
1500
2000
2500
3000
3500
2026-07-08
2026-07-15
2026-07-22
2026-08-01
2026-08-08
2026-08-15
2026-08-22
2026-09-01
2026-09-08
Time
Available Renewable (qAA)
Metered Renewable Generation (qM)
MW
0
1000
2000
3000
4000
2026-07-08
2026-07-15
2026-07-22
2026-08-01
2026-08-08
2026-08-15
2026-08-22
2026-09-01
2026-09-08
Time





https://youtu.be/db4h-TxrIto

1 The shift factor, as it is used today

For a branch e and a generator at node n:

$$\mathrm{SF}_{e,n} = \frac{\Delta f_e}{\Delta P_n}. \tag{1}$$

Two properties define how it is used:

It is transmission-line specific. One number describes one branch against one generator. A constraint group is built by computing a column of these for a single congested circuit.

It is empirical. [WDT] (Appendix 2, p. 71) obtains it by perturbation: move one generator by a fixed amount, re-run the load flow, read the change in branch current. One solve per number.

2 Sensitivity: a new way of looking at it

$$S_{e,n} \;=\; \mathrm{PTDF}_{e,n} \cdot \mathrm{sign}(f_e). \tag{2}$$

The same physical quantity as (1), with the sign re-pointed along each branch's real direction of flow, so that **positive always means relief**.

Without the correction a positive entry means "flow increases in the branch's stored `bus0`→`bus1` direction", which is overload on one branch and relief on another. With it, one row may be ranked, and two rows may be added.

It is **dimensionless**, MW off the branch per MW curtailed at the node, so entries from different branches are directly comparable.

3 How to get to S: K, B, L

K**, incidence,** 754×980**.** Topology only. Column e has exactly two non-zeros: $+1$ at the branch's `bus0`, -1 at its `bus1`.

$$K_{ie} = \begin{cases} +1 & i = \mathrm{bus0}(e) \\ -1 & i = \mathrm{bus1}(e) \\ 0 & \text{otherwise.} \end{cases}$$

B**, susceptance,** 980×980 **diagonal.** Electrical strength only: $B_{ee} = b_e = 1/x_e$, from the per-unit reactance.

$L = KBK^\mathsf{T}$**, the weighted Laplacian,** 754×754**.** Topology and strength combined, and the matrix that must be inverted:

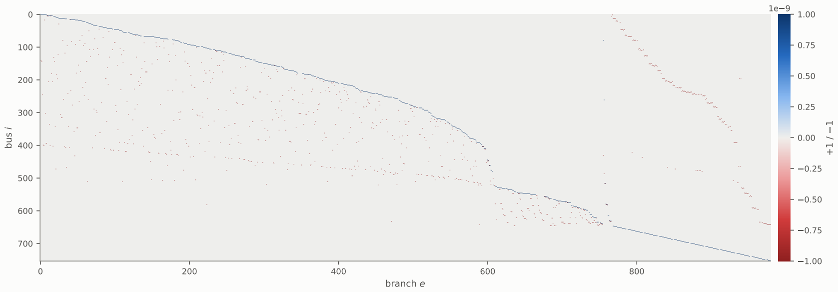
$$L\theta \;=\; p. \tag{3}$$

Why any of this is linear. The AC equations are not. The DC approximation holds bus voltages at 1 p.u., takes $r \ll x$ so losses vanish, and takes angle differences small so $\sin(\theta_i - \theta_j) \approx \theta_i - \theta_j$. Flow is then exactly linear in the angle difference, $f_e = b_e(\theta_i - \theta_j)$. That is the whole reason a closed-form sensitivity exists at all.

4 The matrices themselves

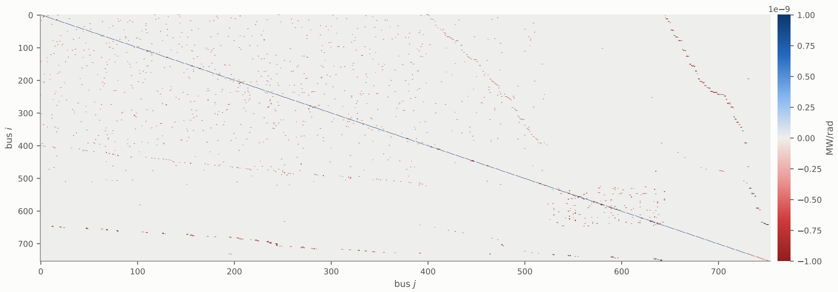
K — **the full 754 × 980 incidence matrix**

Exactly two non-zeros per column: +1 at bus0, −1 at bus1. 1,960 non-zeros, 0.27\% dense.



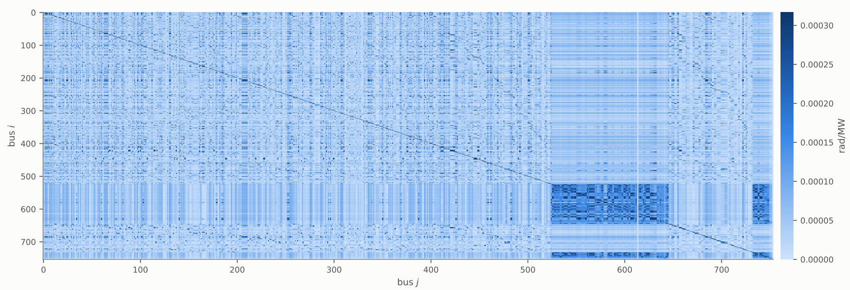
$L = KBK^T$ — **the full 754 × 754 Laplacian**

2,621 non-zeros (0.46% dense). The structure is the grid's own topology.



L^+ — **the full 754 × 754 pseudoinverse**

99.2\% dense. Inversion destroys sparsity: every bus is now coupled to every other.



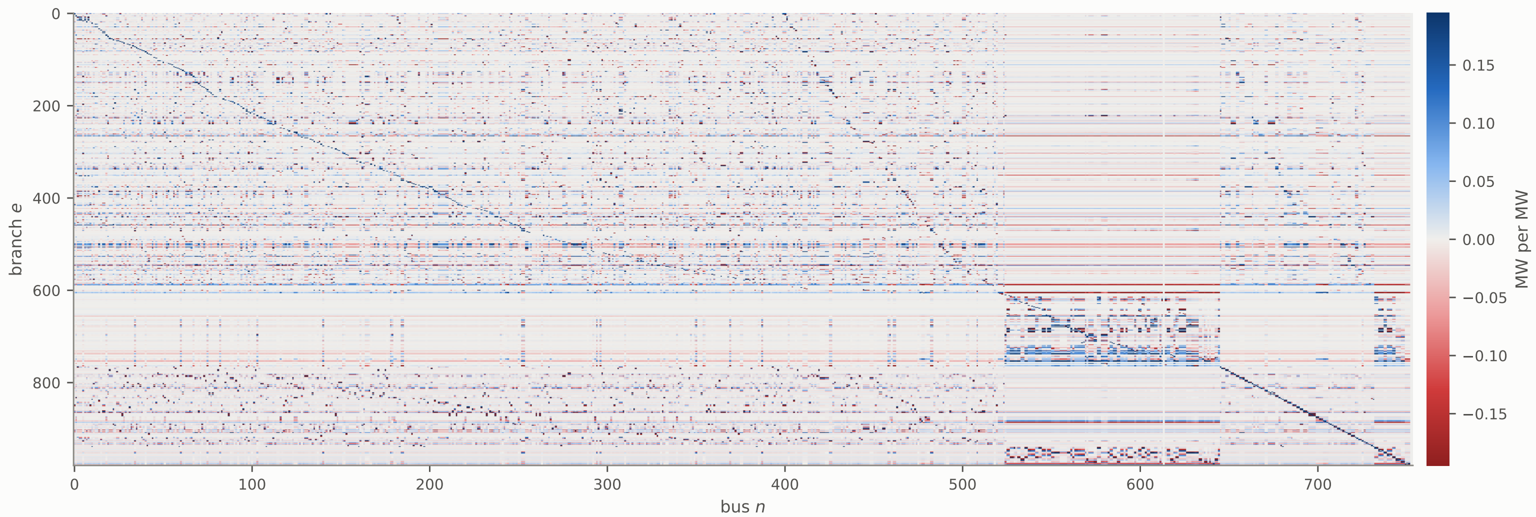
5 The PTDF

$$\mathrm{PTDF} \;=\; BK^{\mathsf{T}}L^{+}, \qquad 980 \times 754, \quad 738{,}920 \text{ entries.} \tag{4}$$

$\mathrm{PTDF}_{e,n}$ is the flow appearing on branch e when 1 MW is injected at bus n and withdrawn across the slack.

PTDF $= BK^{T}L^{+}$ — all 980 × 754 = 738,920 entries

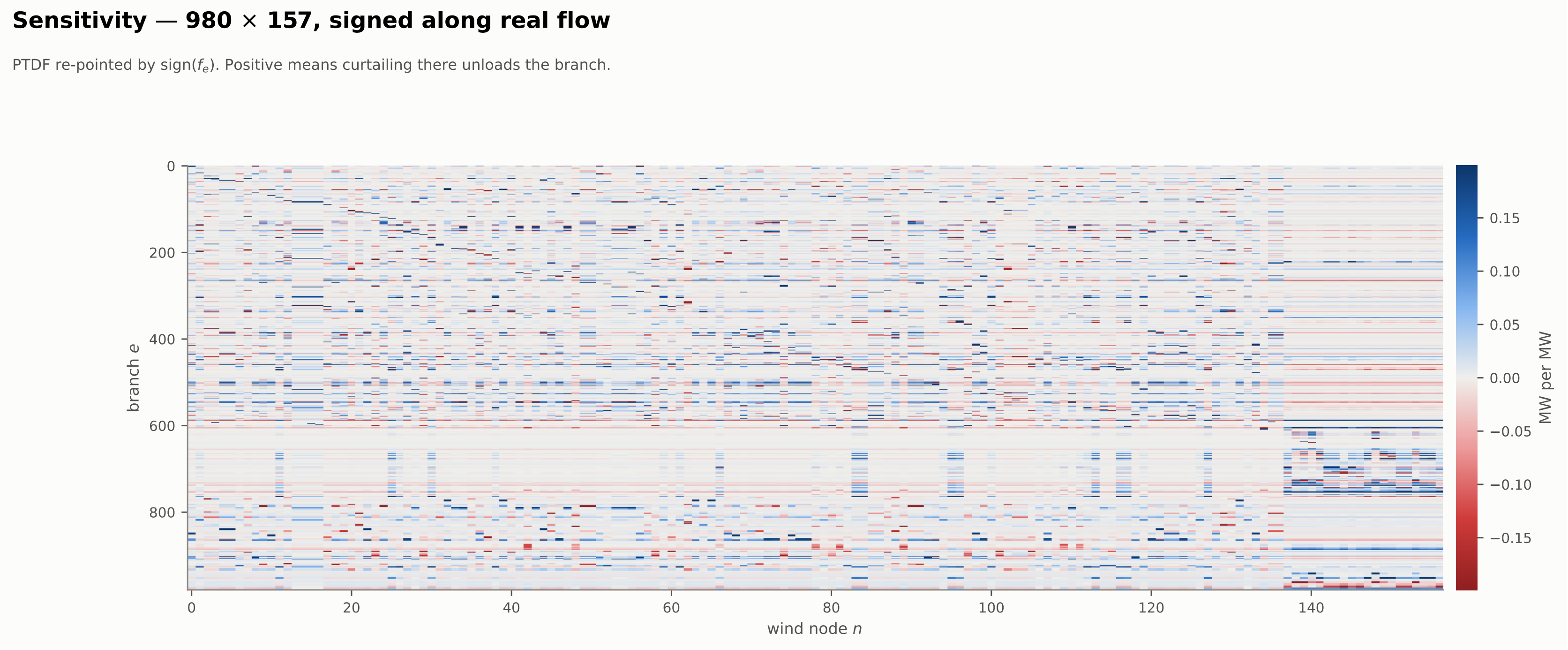
Every branch against every bus. No numeric form of this is readable, so it is shown whole as an image.



6 The sensitivity matrix

Only one quantity in the pipeline moves hour to hour: the direction of flow.

$$p_{\text{eff}} = p + KB\varphi, \qquad f = \text{PTDF} \cdot p_{\text{eff}} - B\varphi, \qquad S_{e,:} = \text{PTDF}_{e,:}\,\text{sign}(f_e). \tag{5}$$

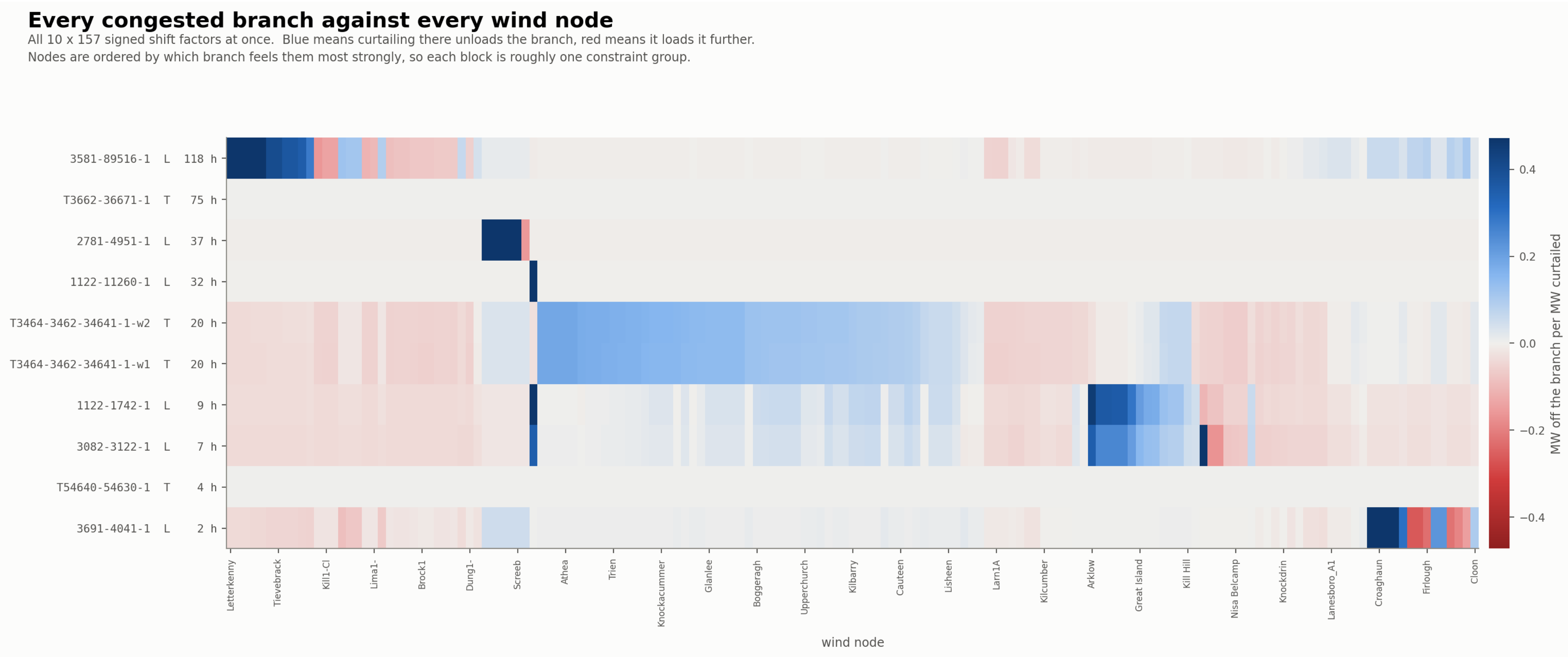


positive	curtailing at that wind node **unloads** the branch
negative	curtailing at that wind node **loads it further**

φ holds the phase shifts of the 2 phase-shifting transformers. Dropping the $-B\varphi$ term puts the flow, and therefore the sign, and therefore every entry of S, wrong on those branches.

7 In practice, only a small set of lines is congested

Of 980 branches, only a handful reach their rating at any hour of the week, and several of those are transformers, which a lines-only model would miss entirely.



Every constraint question in the rest of this document is a question about these rows. The remaining rows exist, are exact, and are never needed until the network changes, which is what an outage is.

8 Net sensitivity: a constraint group that cannot break what it did not look at

The group, as it is drawn today

$$\mathrm{group}(e) \;=\; \{\, n : S_{e,n} \,\ge\, \tau \,\}, \qquad \tau = 0.05. \tag{6}$$

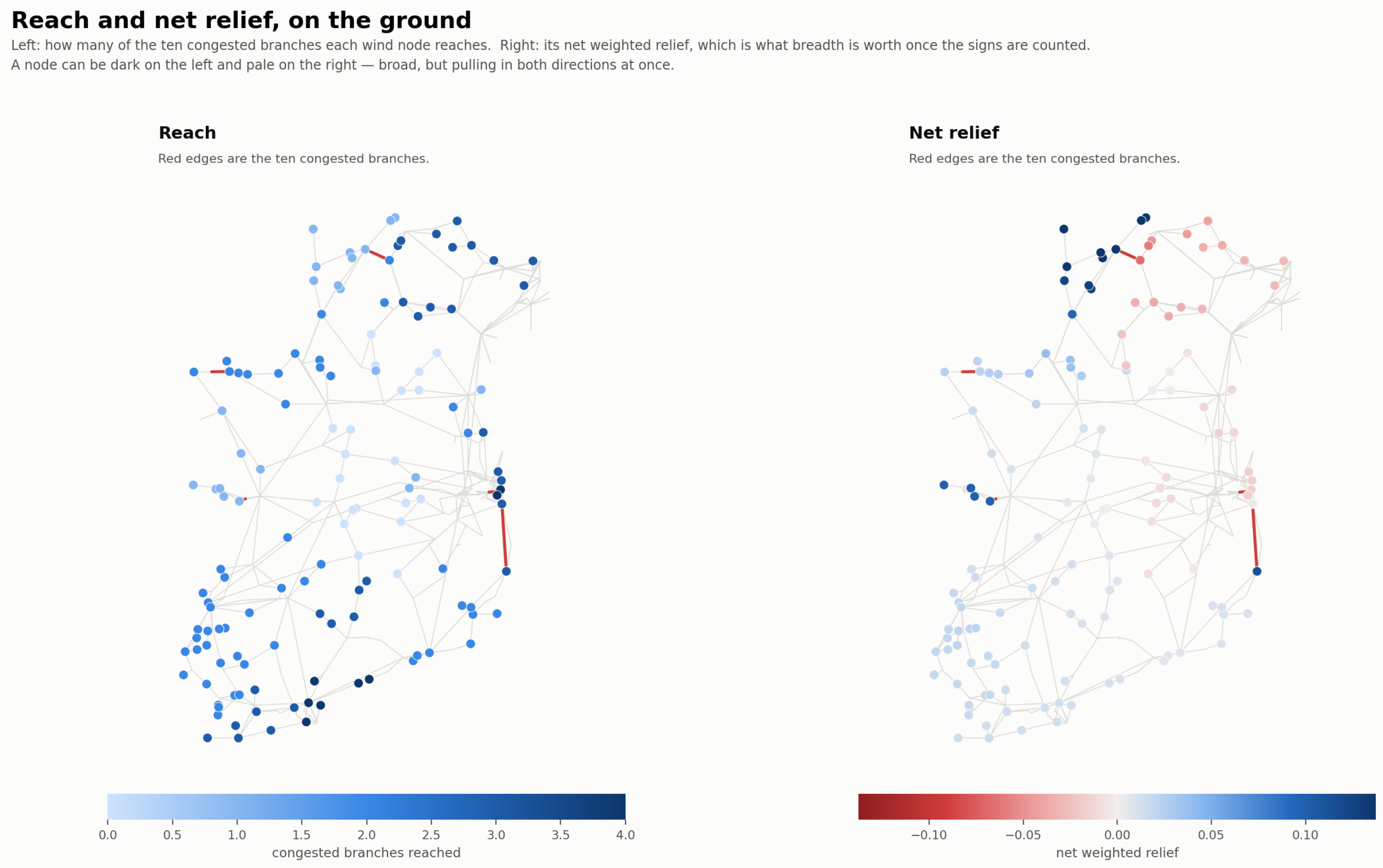
Sensitivities add. Net sensitivity is nothing more than that.

If several branches are over their rating at once, a node's total usefulness is its sensitivities to each of them, summed:

$$S_{\mathrm{net}}(n) \;=\; \sum_{e \,\in\, \mathrm{Over}} S_{e,n}, \qquad \mathrm{Over} = \{\, e : |f_e| > s_e \,\}. \tag{7}$$

The critical principle

No dispatch down of a node may cause an overload on any other branch.



Left: how many congested branches each wind node reaches. Right: its net weighted relief, (7). A node dark on the left and pale on the right is broad but pulling both ways at once, which counting rows cannot distinguish and summing them can.

9 Doing this for every N–1 contingency

The groups are fixed. Membership is read off S, which is topology and reactance only, so it does not move with the hour or the weather. The group for every single-branch outage is built once, offline, and looked up: 432 outages, 58 distinct circuits ever at risk, 1,308 lookup rows.

Most of the fleet is almost never in one. Across all 20,203 events, 119 of 157 nodes, 65.9% of installed capacity, are dispatched down in under 1% of them. The median node keeps 99.61% of everything it offers, and over the whole study only 0.85% of all wind offered is curtailed.

For those generators a constraint group binds on a handful of hours a year. It is not a standing limit on output, and they run at full power essentially all the time.

And a few of these outages do not stop at one circuit.

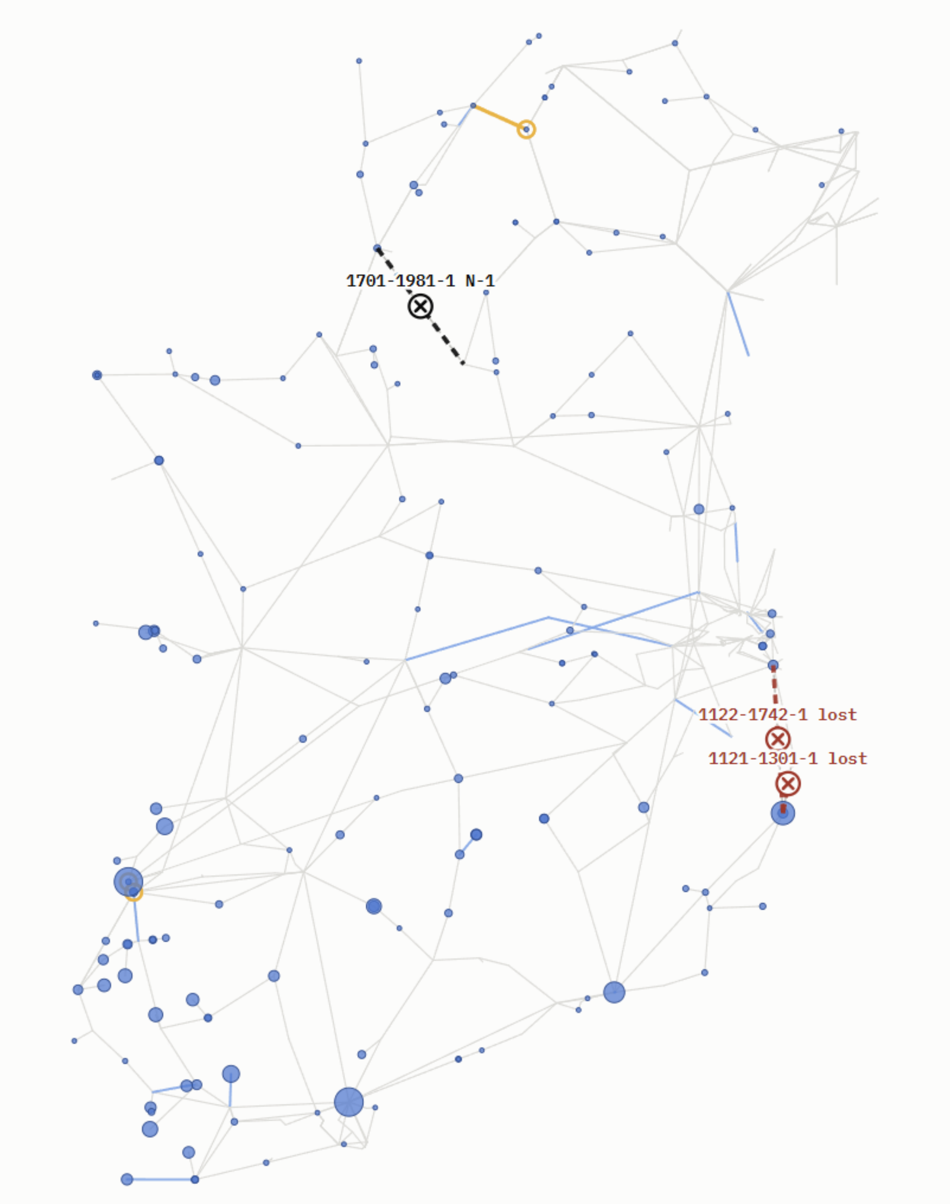
10 Another application: stopping cascading outages

Most single-branch outages are absorbed. Flows redistribute and nothing else reaches its rating. A small number are not: the first trip pushes other elements over their limits, those trip in turn, and the failure propagates.

This is where the two constraint groups separate. A group drawn from one row can relieve the congested circuit while pushing a healthy one over, which starts the next link in the chain. A group built under the critical principle cannot.

Shift factor constraint groups

LINE OUTAGES	WIND CUT
2	744 MW · 14%



Net sensitivity constraint groups

LINE OUTAGES	WIND CUT
0	16 MW · 0%